

PRODUCT REQUIREMENTS DOCUMENT

PulseRank Platform v1.0

Production-simulated marketplace recommendation system.

Field	Value
Version	v1.0
Status	Complete — evidence artifacts generated
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Dashboard	GitHub Pages (public)

1. Problem Statement

Offline recommendation evaluation in marketplace systems fails systematically because click data is position-biased: items shown at rank 1 are far more likely to be clicked than equivalent items at rank 10, regardless of true relevance. Naive NDCG computed from raw click labels severely understates the quality of models that surface good items at lower ranks. PulseRank demonstrates the full correction pipeline — logging display rank at impression time, estimating position propensities, applying IPS debiasing, and making a structured A/B decision against a holdout.

2. Architecture Components

Component	Module	Key Output	Result
Corpus Generation	core/	corpus_summary.json	4,132 sessions, 650 items, 41,320 impressions
Candidate Generation	core/candidates.py	candidate_generation_report.json	Hybrid Recall@100 = 0.690
Ranking Baseline	ranking/baseline.py	ranking_baseline_report.json	NDCG@10 = 0.134 (naive, biased)
IPS Bias Correction	evaluation/ips.py	bias_correction_report.json	IPS NDCG@10 = 0.522
Reranking Constraints	ranking/reranker.py	diversity_report.json	Seller Gini = 0.582
Delayed Attribution	simulation/attribution.py	conversion_attribution_report.json	30-day window, attributed labels
Offline A/B Simulation	simulation/ab.py	ab_simulation_results.json	HOLD_SIMULATED
Failure Scenarios	artifacts/failures.py	failure_recovery_report.json	15 scenarios documented

3. Failure Scenarios

Cold-start: new items

New items have popularity score = 0 and no content features. They rank last. The recovery path is a synthetic popularity floor (minimum exposure boost for first 48h) and content-based fallback using item metadata similarity.

Acceptance: cold_start_item section in failure_recovery_report.json documents the fallback trigger condition and estimated exposure share.

Position bias without IPS correction

Using naive NDCG on position-biased click data produces $NDCG@10 = 0.134$. The model appears poor. With IPS correction, $NDCG@10 = 0.522$ — the model quality was being masked by display-rank effects.

Acceptance: bias_correction_report.json shows pre/post IPS delta and clipped weight distribution.

Propensity estimator misconfiguration

If the click model underestimates propensity at rank 1, IPS over-weights rank-1 items and inflates the metric. Detection: propensity_by_rank_report.json must show monotonically decreasing propensity from rank 1 to rank 10.

Acceptance: propensity_by_rank_report.json: $\text{propensity}[\text{rank}=1] > \text{propensity}[\text{rank}=5] > \text{propensity}[\text{rank}=10]$ strictly.

Gini constraint violation

A seller captures >40% of top-10 impressions before reranking. The reranker's seller constraint applies a cap and redistributes slots. Gini drops from pre-rerank value to 0.582 post-rerank.

Acceptance: diversity_guardrail_log.json records per-session constraint decisions and Gini delta.

Attribution window right-censoring

Purchases after 30 days are unattributed. For high-consideration items (luxury goods), this may undercount conversions. The mitigation is to report coverage at 1d, 7d, 30d in conversion_attribution_report.json.

Acceptance: conversion_attribution_report.json: attribution_coverage_30d captures $\geq 85\%$ of all purchases in the synthetic corpus.

4. Quantitative Acceptance Criteria

Criterion	Target	Evidence Artifact
Hybrid Recall@100	≥ 0.65	candidate_generation_report.json
IPS NDCG@10 > naive NDCG@10	IPS/naive ratio ≥ 2.0	bias_correction_report.json
Seller Gini post-rerank	< pre-rerank Gini	diversity_report.json
Attribution coverage at 30d	$\geq 80\%$ of purchases	conversion_attribution_report.json
A/B decision label	HOLD_SIMULATED or LAUNCH_SIMULATED	hold_simulated_results.json
JSON artifacts count	≥ 30	outputs/evidence/
Failure scenarios documented	≥ 15	failure_recovery_report.json
Display-rank coverage	1.0 (all impressions have rank)	corpus_summary.json

5. Truth Boundary

PulseRank is production-simulated. It does not claim real users, real revenue, live A/B testing, online inference, RL exploration, or production deployment. Every claim is backed by an executable script and a JSON artifact. The HOLD_SIMULATED label explicitly distinguishes offline evidence from a live experiment result.